

Lab 2: Exploratory Data Analysis

T-504-ITML - Machine Learning / T-711-FOML - Fundamentals of Machine Learning

Fall, 2026

1 Introduction

The goal of this lab is to get accustomed to working with data in Python and exploring data through visualization using Python. You will need the pandas and seaborn libraries. You can work in groups of up to three students (ideally 2).

2 Handing in

Hand in a pdf report with the answers to all the questions below. For each answer also show the Python code that generated the answer or the plot. A PDF export of your Colab or Jupyter notebook should be fine, if it is clear which question you are answering.

3 Tasks

1. (*Total points: 20*) Write a Python program to analyze the Iris dataset. You will find the Iris.csv file on Canvas. You can load the dataset in Python using pandas with

```
import pandas as pd
iris = pd.read_csv("Iris.csv")
```

This reads the csv file into a pandas DataFrame object. A DataFrame is an easily accessible and modifiable representation of tabular data. Use the DataFrame to let your program print answers to the following questions:

- (a) (2 points) How many instances (examples) are in the dataset?
 - (b) (5 points) How many attributes (columns) are there? What are their names?
 - (c) (8 points) For each of the attributes: What is the range of values (minimum and maximum)? What is the average, standard deviation and median of the values?
 - (d) (5 points) What are the possible output names (species)?
2. (*Total points: 30*) Use seaborn to explore the data visually using box plots and scatter plots. Especially the functions boxplot and pairplot may be useful.
 - (a) (10 points) Which one of the attributes has the largest/smallest variance? Make a plot that easily shows the answer.
 - (b) (10 points) Does the variance of the attributes depend on the species? Make a plot that easily shows the answer.
 - (c) (10 points) Which of the three species can be easily (linearly) separated from the remaining two? Write down a simple rule of the form "If attribute $X < | > V$ then class = Y else class $\neq Y$ ", that is, pick an attribute and a threshold value that best separates one class from the others. Generate a plot that supports your rule.
 3. (*Total points: 50*) Use similar methods as above to analyze a dataset of passengers of the Titanic (titanic.csv on Canvas). The dataset contains some information about the passengers of the Titanic, such as age, sex, class and price of the ticket, number of family members on board, and most importantly, whether they survived or not. The dataset is a slightly processed version of the dataset

that is described here, the extra processing (which added a few attributes) was done with this script. You can also find the raw data, in case you prefer to use this. The dataset is real world data which is incomplete (missing attribute values) and sometimes misleading (e.g., ticket prices are not the price per passenger, but the total price for all passengers on the same ticket). The dataset has some numeric and some nominal attributes. Different graphs work differently well depending on the type of attribute used in the graph.

- (a) (20 points) Explore the effects of the "woman and children first" policy on the survival rate. (The "who" attribute might come in handy for this.) What can be said about the effects of that policy depending on the class of the passenger? Generate one or more graphs that support your conclusions.
- (b) (30 points) Generate at least two graphs of different type (box plot, scatter plot, histogram, bar plot, line plot) that show some interesting facts about the passengers and describe what can be seen in the graphs. (For example, you could look at the distribution of attribute values, correlations between different attributes, or correlations between pairs of attributes and survival.)